

CALIBRATE THE BENCHMARK BEFORE RANKING THE MODELS: AN OPEN, AUDITABLE MIR STACK FOR ELECTRONIC DANCE MUSIC

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ABSTRACT

`jams` is an open-source analysis service for electronic dance music: key, tempo, beats, functional structure, stems, and per-stem MIDI from one HTTP call, with every reported number regenerated by committed code. Rebuilding standard benchmarks under a single paired-statistics protocol surfaced a calibration problem that affects how key detection results are compared. The strongest published GiantSteps Key system gains +0.087 weighted score when re-run on our 567-track usable subset, six times the field’s total claimed progress since 2018, so subset-measured scores cannot be compared against published full-set scores at all. Under same-subset paired tests, a 109k-parameter CNN trained on public Beatport annotations ties that system ($\Delta = -0.001$, 95% CI $[-0.019, +0.018]$) with better exact accuracy and MIT-licensed weights. We demonstrate the service and its companion web application (waveform, beat grid, segment editor, per-stem piano rolls) running locally on a laptop.

1. INTRODUCTION

DJ and production workflows ask a small set of questions about any track: its key, its tempo, where the beats and phrase boundaries fall, and what the constituent parts are. EDM is the genre family where these questions matter most operationally and where general-purpose MIR models underperform, since half-time drum & bass, modally ambiguous pads, and drops that behave nothing like pop choruses sit in the tails of most training distributions.

`jams` packages an EDM-focused answer as a local service (macOS and Linux) exposing a single endpoint.¹ Analysis capabilities and their held-out results: key (0.832 MIREX weighted, §2), tempo (0.967 Acc1 on corrected labels, $n=458$), beats and structure via an on-device All-In-One ensemble [1, 2] (0.978 beat-F), SCNet separation [3], and a mix-to-MIDI cascade whose accompaniment

¹<https://github.com/jhurliman/jams> (MIT). The repository contains the service, acquisition and scoring scripts for every dataset used, an experiments ledger that every claim traces to, and the statistics script that regenerates every confidence interval reported here.



transcription (0.788 note-F on Slakh2100 [4]) exceeds the *ground-truth-stem* ceiling of the widely deployed basic-pitch transcriber (0.490) [5, 6]. Everything learned that ships is MIT-licensed, trained by us on public data.

This abstract focuses on the finding we most want the community to poke at before the full paper: benchmark-subset calibration in key detection.

2. THE SUBSET-CALIBRATION FINDING

GiantSteps Key [7] ships 604 annotations. In our pull, 600 have retrievable preview audio; like prior work we drop atonal, multi-key, and unparseable labels, leaving $n=567$. Such subsets are routine and rarely identical across papers. To measure what subset selection alone does to scores, we re-ran the strongest published system, the madmom key CNN [8], on our subset with its published weights and default settings. Published full-set score: 0.746 weighted. On our subset: **0.833**, a +0.087 shift from track selection alone (Figure 1).

For scale, total claimed progress across published GiantSteps Key systems from 2018 to 2026 (0.746 to 0.759 [9]) is 0.013. The subset effect is more than six times larger than eight years of reported improvement. Any comparison of a subset-measured number with a published full-set number is therefore uninterpretable, including comparisons that favor our own systems. The only rankings we trust are same-subset paired tests, and we publish the subset track list and per-track outputs so future systems can be compared on exactly these terms.

3. PARITY WITHOUT THE LICENSE RESTRICTION

Under that protocol we evaluated a deliberately small key CNN of our own: log-CQT input, three convolutional blocks, time-only average pooling so frequency structure reaches the readout, and a 24-way softmax; 109k parameters, 0.45 MB. It trains in two hours on `beatport_key`, a revised-annotation superset of GiantSteps-MTG-Keys [10] (1,363 usable tracks, zero track-id overlap with the test set). The experiment was pre-registered in the public ledger with a fixed decision rule; all model selection used training-corpus cross-validation, and the locked model received exactly one test evaluation.

The result (Table 1) is a statistical dead heat with the strongest published system: paired $\Delta = -0.001$ $[-0.019,$

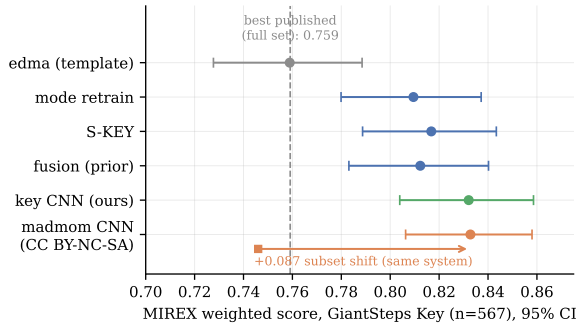


Figure 1. Key systems on the same $n=567$ subset with 95% bootstrap CIs. The square/arrow shows the *same* madmom CNN’s published full-set score (0.746) against its score on our subset (0.833): a +0.087 shift from subset selection alone, larger than the spread between the five strongest systems.

system	weighted	exact	license
edma template [11]	0.759	0.688	AGPL
S-KEY [12]	0.817	0.748	MIT
key CNN (ours)	0.832	0.780	MIT
madmom CNN [8]	0.833	0.773	CC BY-NC-SA

Table 1. GiantSteps Key, $n=567$ usable subset, MIREX weighted and exact accuracy. Same-subset scores only; full CIs in the repository.

4. FINDINGS THAT TRAVEL

The same harness produced measurement results that changed our numbers by more than typical model improvements do. Slakh bass MIDI follows written convention: scoring sounding-pitch output against it without the +12 semitone correction turns 0.80 note-F into 0.04. The 16 kHz preview build of Slakh caps drum onset-F near 0.46 versus 0.64 at full bandwidth. GiantSteps tempo labels are wrong for half-time genres; correcting them from the v2 re-annotation [13] moves our tempo stack from 0.919 to 0.967 Acc1, a larger effect than any model change we measured. In separator selection, SCNet’s +2.7 dB SI-SDR gain on drums over HT-Demucs costs one point of downstream drum onset-F, so we score separators on the task they feed. And three pre-registered variants of class-rebalanced structure training all failed their gates, every per-class gain funded by an adjacent-class regression; the negative result ships in the ledger with the same statistics as the positives.

5. DEMO

The demo is the full stack live on a laptop: drop in a track, get key, tempo, beat and downbeat grid, functional segments (intro/buildup/drop/breakdown), stems, and per-stem MIDI in one response, then inspect everything in the companion web application, which overlays the beat grid and segment boundaries on the waveform and renders each stem’s transcription as a piano roll. The same interface is the label-inspection tool behind our error analyses, so visitors can see exactly how the reported numbers were audited, per track, against the ledger.

6. AI USAGE STATEMENT

Experiments, infrastructure, and drafting were carried out with substantial assistance from Claude (Anthropic), operating the evaluation harness described here under the author’s direction. All quantitative claims regenerate from committed code, and the pre-registered gates and one-shot test evaluations described above bound what any contributor, human or machine, could tune against the test sets.

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